Single RNN layer model :

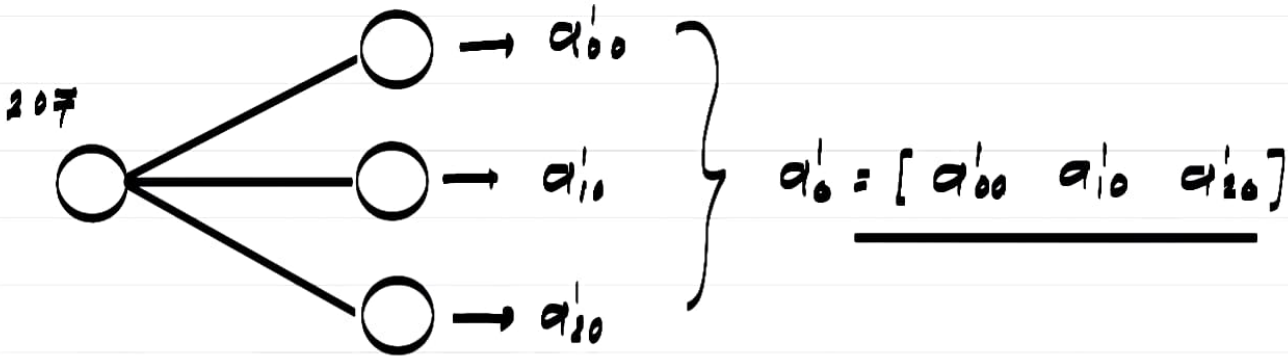
The movie is really good — 5 tokens

↓ tokenization

[207, 301, 402, 500, 305] — 5 token ids

a_0^0 a_1^0 a_2^0 a_3^0 a_4^0

at $t=0$



$z_{00}^1 = w_{00}^1 (207)$ $\quad a_{00}^1 = f(z_{00}^1)$

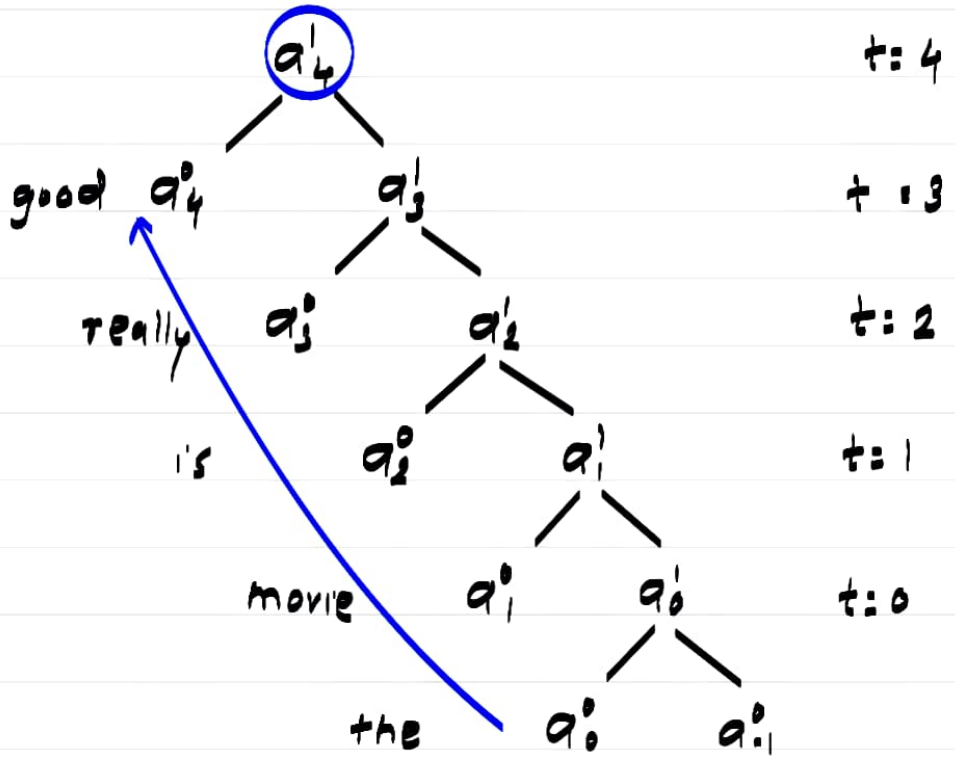
$z_{10}^1 = w_{10}^1 (207)$ $\quad a_{10}^1 = f(z_{10}^1)$

$z_{20}^1 = w_{20}^1 (207)$ $\quad a_{20}^1 = f(z_{20}^1)$

w_{ij}^l — l - layer, i - node i in layer l, j - node j in layer $l-1$

z_{ij}^l — l - layer, i - node i in layer l, j - j^{th} token in sequence

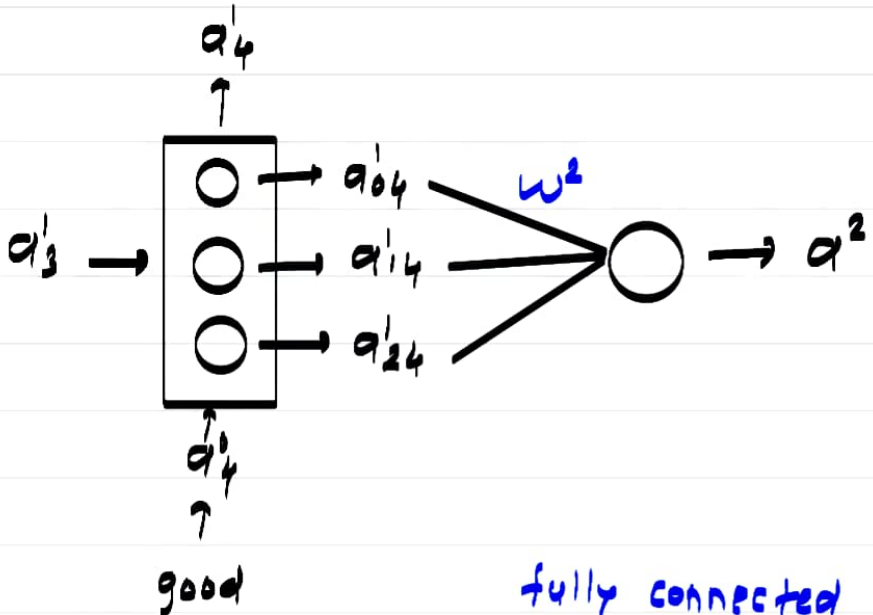
$z_0^1 = w^1 a_0^0$



a^1_4 holds the information of all tokens and also knows the order of tokens

as there are 3 nodes in the RNN layer

$a^1_4 : [\, a^1_{04} \quad a^1_{14} \quad a^1_{24} \,]$



$z^2 = w^2 a^1_4$

$a^2 = \text{sigmoid}(z^2)$

as its a binary clf +ve/-ve

The movie is really good 5 tokens

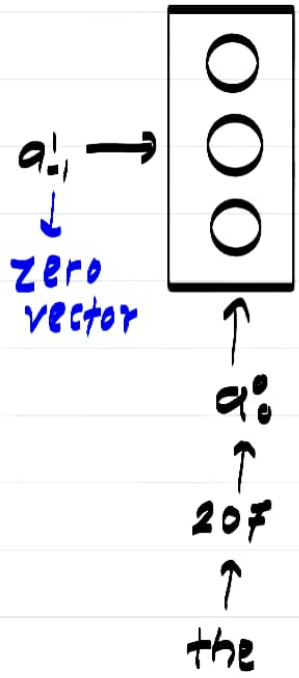
↓ tokenization

[207, 301, 402, 500, 305] 5 token ids

a_0^0 a_1^0 a_2^0 a_3^0 a_4^0

t : 0 1 2 3 4

$a_0^1 = [a_{00}^1, a_{10}^1, a_{20}^1]$



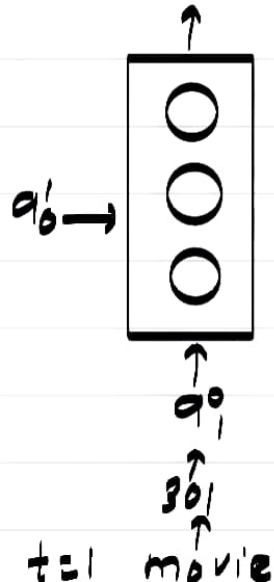
$z_0^1 = w^1 a_0^0 + w_a a_{-1}^0$

$z_0^1 = w^1 a_0^0 + 0$

$a_0^1 = f(z_0^1)$

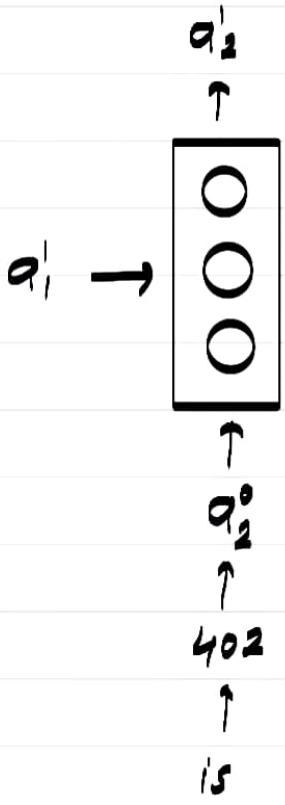
t : 0

$a_1^1 = [a_{01}^1, a_{11}^1, a_{21}^1]$



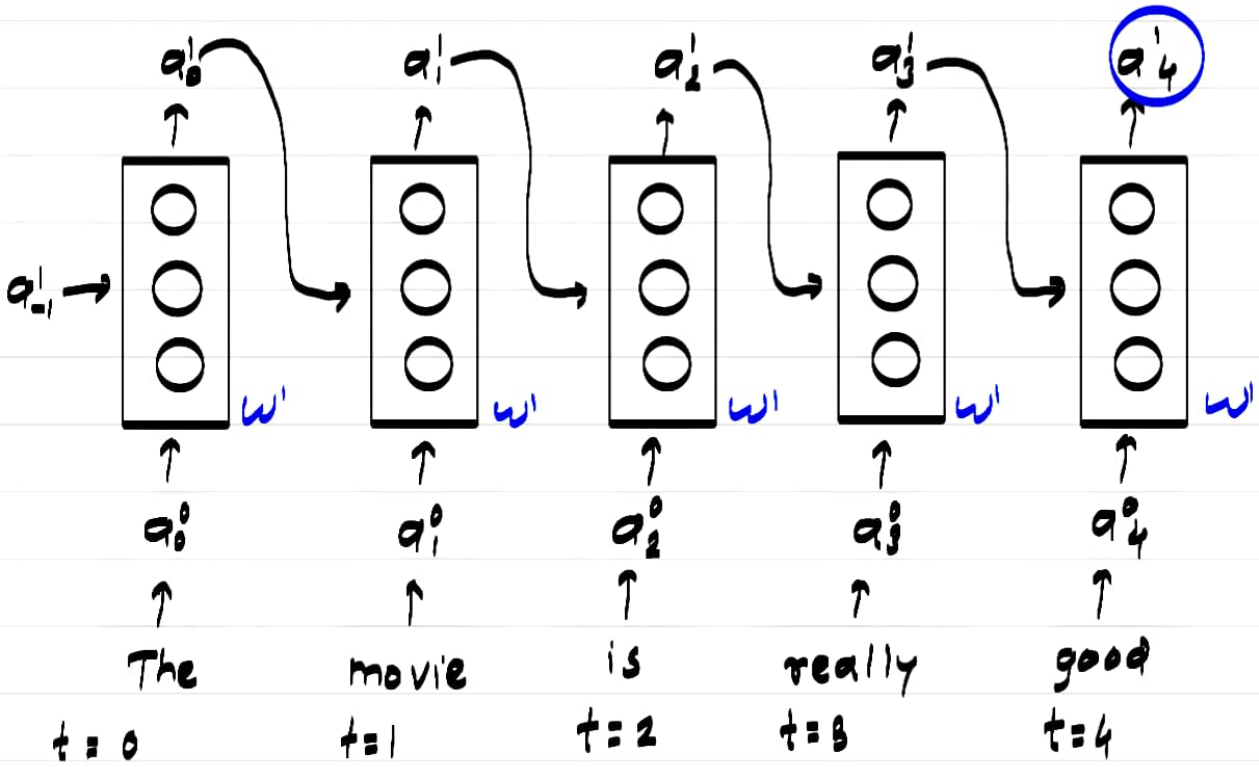
$z_1^1 = w^1 a_1^0 + w_a a_0^1$

$a_1^1 = f(z_1^1)$



$$z^1_2 = w^1 a^0_2 + w_a a^1_1$$

$$a^1_2 = f(z^1_2)$$



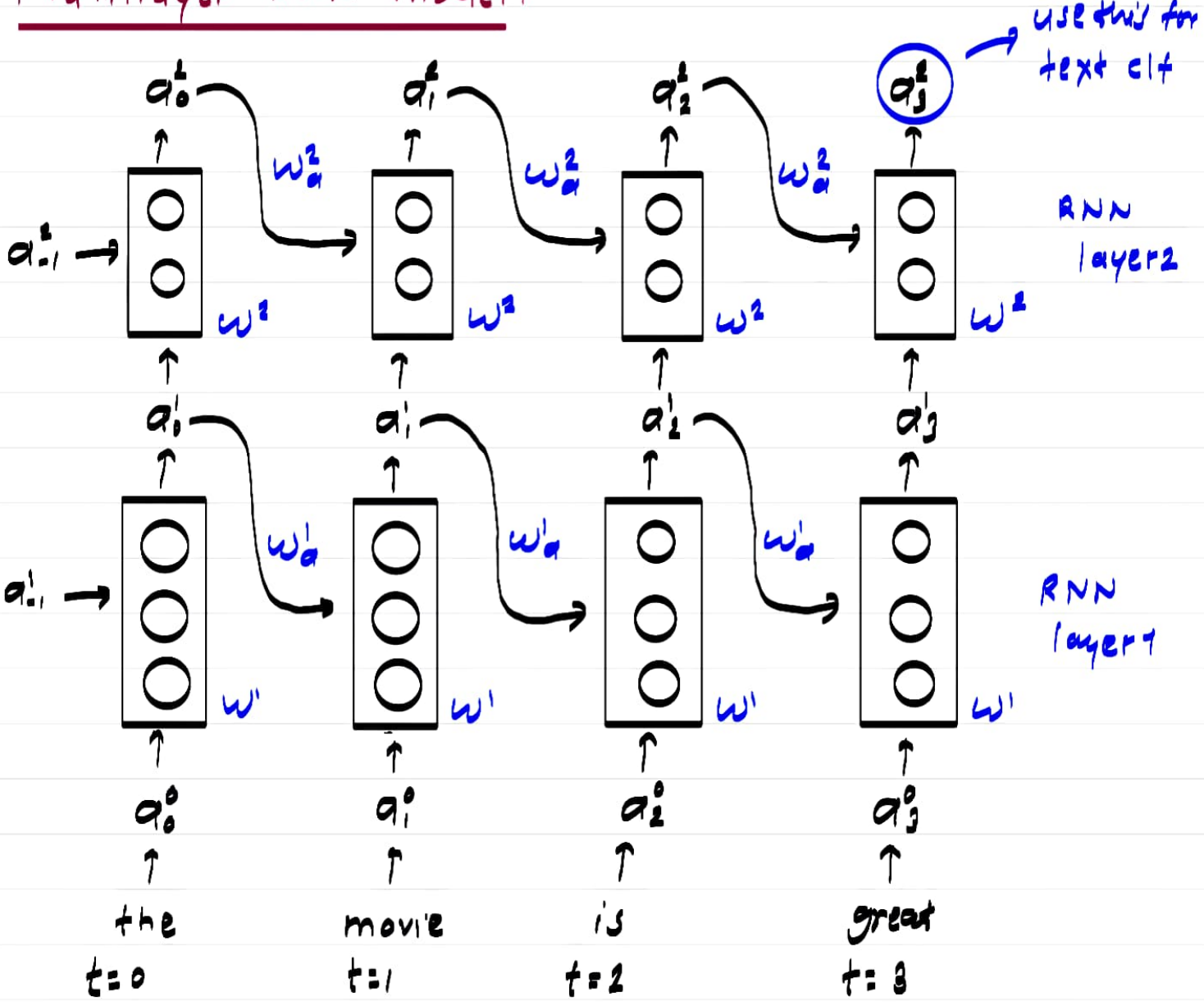
at $t = i$

$$z^1_i = w^1 a^0_i + w_a a^0_{i-1}$$

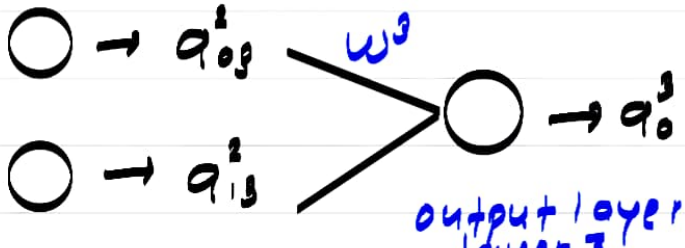
$$a^1_i = f(z^1_i)$$

The RNN layer is used in multiple time-stamps with same weights

Multilayer RNN model:



$a^2_3 = [a^2_{03} \quad a^2_{13}]$ 2 nodes in RNN layer 2



$z^3_0 = w^3 a^2_3$

$a^3_0 = \text{sigmoid}(z^3_0)$

Token classification model using RNN:

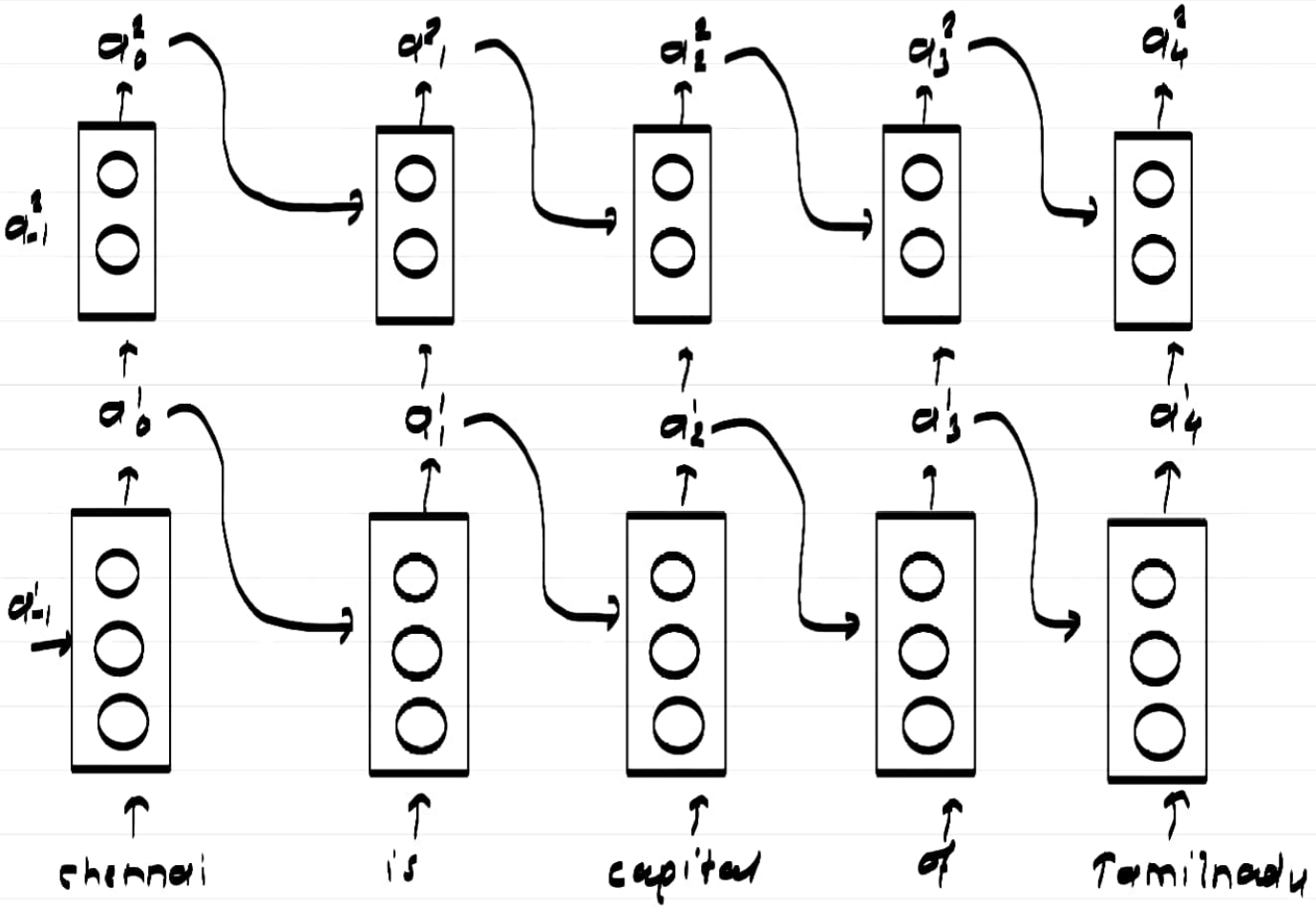
eg.

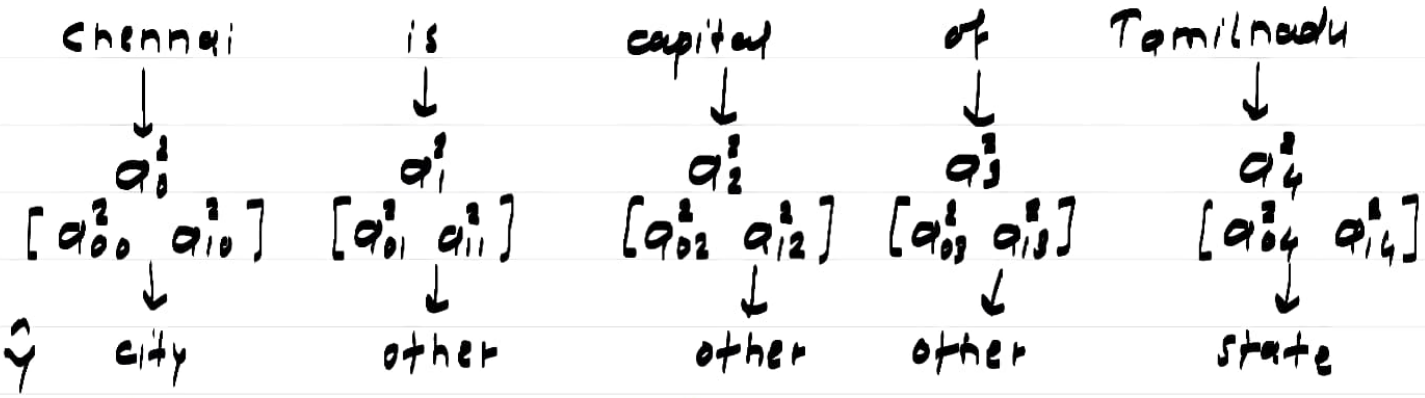
text1 :	New	Delhi	is	capital	of	India
labels :	city	city	other	other	other	country

text2 :	Chennai	is	capital	of	Tamilnadu
labels :	city	other	other	other	state

unique labels: city country state other

4 labels





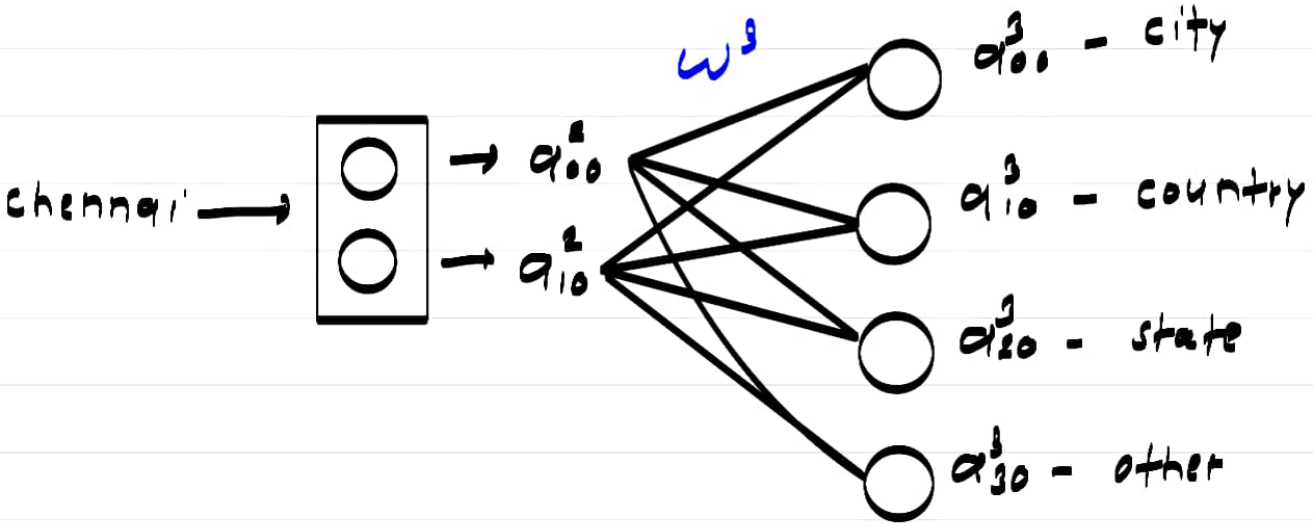
Labels : city country state other

this will be a multiclass clf model

↓

softmax activation in output layer

no. of nodes in output layer = no. of unique classes



$$z^3 = w^3 a^2$$

$$a^3 = \text{softmax}(z^3)$$

a_{00}^3 - p(city) ✓

a_{10}^3 - p(country)

a_{20}^3 - p(state)

a_{30}^3 - p(other)

} maximum probability class will be the predicted token type